The representativeness map: every failure mode we measure on the VCS has a documented neural-network counterpart

The single strongest objection — "a deterministic 1977 chip cannot stand in for a learned, distributed, stochastic neural network" — answered as a necessary-condition screen.

The asymmetry is the argument: on the VCS each failure is PROVABLE — exact intervention oracle, full observability, no learning; on neural networks the same failure is only SUSPECTED. A method that fails here has no business being trusted there (necessary, not sufficient).

Failure mode observed & MEASURED on the VCS	**VCS evidence (PROVABLE)** scored vs the exact §1 oracle		**Neural-network counterpart** documented analogue (cite-ready)	**Status** here → there
1 **Gradient vanishes on discrete / index outputs**	**0.000** position-regime F	Vanilla saliency & IG faithfulness = 0.000 on the position regime; whole gradient bucket = 0.068 (n=9). Provably 0 (strobe-timed argmax, §1).	Shattered / saturated gradients; non-differentiable argmax & index readouts. [Balduzzi 2017; Sundararajan 2017 (IG motivation)]	PROVABLE here → suspected there
2 **Saliency is model-invariant (Adebayo sanity-check fails)**	**0.000** sanity-check margin	Program-randomization test: saliency cosine = 1.000 while 95% of RAM changed → sanity check FAILS. Constant Jacobian, no ReLU.	Saliency unchanged under model- & label-randomization; looks like an edge detector. [Adebayo et al. 2018, NeurIPS]	PROVABLE here → suspected there
3 **Correlation ≠ causation (Granger & tuning trap)**	**0.136** Granger F	Granger edges faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116 (spurious tuning) — vs exact oracle.	Spurious Granger edges; 'present ≠ used' tuning & attention as explanation. [Anand 2019; Jain & Wallace 2019 (attention)]	PROVABLE here → suspected there
4 **Single-unit ablation has an interaction blind-spot**	**0.414** connectome F	Single-unit lesion importance is faithful (0.987) yet MISSES the wiring: connectome graph recovered at only 0.414; SAE feature ablations move y in 0 of the audited cells.	Single-unit ablation misleading; computation is distributed across units. [Morcos et al. 2018; Leavitt & Morcos 2020]	PROVABLE here → suspected there
5 **SAE / probe: present-not-used + polysemanticity**	**0.162** selectivity F	SAEs match a known var (100% of candidates in 4/6 games, min 83%) yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are decodable-not-causal → selectivity F = 0.162.	Control-task selectivity gap; superposition / polysemantic neurons (features in superposition). [Hewitt & Liang 2019; Elhage et al. 2022]	PROVABLE here → suspected there
6 **Baseline-sensitivity of Integrated / Expected Gradients**	**0.034** EG F (all-regime)	IG magnitude is baseline-sensitive (corr_sensitivity up to 0.999 across games); a baseline = the content byte kills IG; EG lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input. [Kindermans et al. 2019; Sturmfels et al. 2020]	PROVABLE here → suspected there

Headline (position regime): causal/intervention 0.412 vs gradient/correlational 0.068 — a 0.344 faithfulness gap; activation patching = 1.000 (oracle ceiling) where vanilla saliency = 0.000 (naive-gradient floor).

- VCS evidence — measured vs the exact §1 intervention oracle
- NN counterpart — documented analogue (conceptual mapping, cite-ready)
- PROVABLE on the VCS (ground truth)
- only SUSPECTED on neural nets

All VCS numbers read from committed records: leaderboard.json (P2-E6-1), faithful_demo.json (P2-E6-3), guided_backprop_core_summary (Adebayo), ig_baseline_sweep_core_summary, sae_core_summary, linear_probing_core_summary. No experiment re-run. NN-counterpart labels per experiment_design.md §4–§7 and plan.md (the representativeness rebuttal).